Software Manual (Generated from the source code with the help of ChatGPT)

Automated Ruby Fluorescence Preprocessing and Dual-Voigt Line Fitting Pipeline

Overview

This software provides a complete batch-processing workflow for ruby fluorescence spectra acquired using an Avantes spectrometer. The pipeline imports raw text files, performs baseline correction and normalization, automatically identifies the R1 and R2 emission lines, fits both peaks simultaneously using constrained Voigt functions, evaluates fit quality, computes derived spectral parameters, and exports publication-ready datasets.

Detailed Overview

This software implements a complete batch-processing workflow for the analysis of ruby fluorescence spectra acquired during high-pressure experiments performed in a Diamond Anvil Cell (DAC). The primary objective of the workflow is to determine the spectral positions and linewidths of the ruby R_1 and R_2 emission lines through robust, automated Voigt-profile fitting, providing accurate and reproducible spectral measurements across large experimental datasets.

Within DAC experiments, the pressure experienced by the sample is routinely determined from the pressure-dependent wavelength shift of the ruby R_1 fluorescence line. Accurate determination of the R_1 peak position is therefore essential for reliable pressure calibration. Simultaneously, the spectral separation between the R_1 and R_2 lines and the linewidth of the R_1 transition provide valuable indicators of pressure gradients and non-hydrostatic stress within the pressure medium.

Rather than calculating pressure directly, this software focuses on the reproducible extraction of the underlying spectral parameters required for subsequent pressure analysis. For each spectrum, the workflow performs automated baseline correction, intensity normalization, peak detection, constrained dual-Voigt fitting of the R_1 and R_2 emission lines, quality assessment, and statistical summarization. The resulting peak positions, linewidths, fitting statistics, and uncertainty-related metrics are exported in formats suitable for further processing.

The exported datasets are subsequently imported into OriginPro, where the fitted R_1 wavelength is converted into ground-truth pressure values using the appropriate ruby pressure calibration equation. OriginPro is also used to propagate experimental uncertainties, calculate pressure repeatability, and evaluate hydrostaticity through analysis of the R_1–R_2 peak separation and R_1 linewidth. By separating spectral fitting from pressure calculation, the workflow maintains a clear distinction between experimental data processing and calibration, allowing pressure scales or uncertainty models to be updated without modifying the underlying spectral analysis software.

The software has been designed for automated batch processing of large collections of ruby fluorescence spectra, providing a reproducible and standardized methodology suitable for high-pressure spectroscopy experiments, pressure calibration studies, and routine Diamond Anvil Cell measurements.

Software Architecture

```
avantes_data_loader.py
   │
   └── load_folder()
         └── load_single_file()

preprocess_spectra_and_fit_ruby_lines.py
   ├── preprocess_single_spectrum()
   ├── build_init_and_bounds()
   ├── select_fit_region()
   ├── two_voigt_model()
   ├── fwhm_voigt_approx()
   ├── export_per_spectrum_table()
   ├── write_results_tsv()
   ├── write_for_origin_txt()
   ├── plot_diagnostic()
   └── main()
```

Module: avantes_data_loader.py

Purpose

Provides robust import of Avantes Rwd8 text exports for batch analysis.

Functions

load_folder()

Searches a directory for Avantes .txt exports, validates the folder, loads every spectrum via load_single_file(), and returns a dictionary keyed by filename.

load_single_file()

Parses the exported text file, automatically locates the spectral table, extracts wavelength (column 1) and intensity (column 5), skips headers and malformed rows, optionally truncates the dataset, and returns NumPy arrays.

Module: preprocess_spectra_and_fit_ruby_lines.py

Purpose

Implements the complete preprocessing, fitting, quality-control and export workflow.

Functions

fwhm_voigt_approx()

Calculates an approximate Voigt full-width-at-half-maximum from Gaussian and Lorentzian width parameters.

two_voigt_model()

Defines the nonlinear model consisting of a constant baseline plus independent Voigt profiles representing the R1 and R2 ruby emission lines.

preprocess_single_spectrum()

Performs baseline estimation from the spectral tail, subtracts the baseline, normalizes the spectrum, detects the R1 peak, predicts the R2 location, searches for the secondary peak, and returns all preprocessing metadata required for fitting.

build_init_and_bounds()

Constructs physically constrained initial parameter estimates and optimisation bounds for scipy curve_fit including independent centres, amplitudes, Gaussian widths and Lorentzian widths.

select_fit_region()

Automatically extracts only the wavelength interval surrounding the ruby doublet, improving fitting stability and computational efficiency.

export_per_spectrum_table()

Exports wavelength, normalized spectrum, fitted spectrum and residuals for each individual spectrum.

write_results_tsv()

Generates a batch summary containing fitted peak positions, widths, amplitudes, RMSE and preprocessing statistics.

write_for_origin_txt()

Produces an Origin-compatible summary table containing mean spectral parameters, repeatability statistics and placeholders for pressure and temperature uncertainty calculations.

plot_diagnostic()

Displays optional diagnostic figures illustrating preprocessing, peak detection and the final Voigt fit.

main()

Coordinates the complete workflow: imports spectra, preprocesses data, performs constrained nonlinear fitting, evaluates fit quality, computes derived parameters, exports results, generates summaries and reports processing statistics.

Processing Workflow

- Batch import of Avantes spectra.
- Automatic baseline estimation using a user-defined spectral tail.
- Baseline subtraction.
- Intensity normalization.
- Automatic detection of the R1 emission peak.
- Prediction and detection of the R2 peak.
- Construction of constrained optimisation parameters.
- Automatic selection of the fitting window.
- Simultaneous dual-Voigt nonlinear least-squares fitting.
- Calculation of FWHM, peak separation and RMSE.
- Validation of physically meaningful fits.
- Generation of diagnostic plots.
- Export of individual fits, summary tables and Origin-ready datasets.

Output Files

- Per-spectrum wavelength/intensity/fit/residual tables
- Batch TSV summary
- Batch TXT summary
- Origin-compatible summary file
- Optional diagnostic figures
- Console processing statistics

Quality Control

The workflow validates spectrum dimensions, verifies baseline indices, rejects invalid spectra, uses bounded nonlinear optimisation, evaluates RMSE, checks fitted peak ordering and width positivity, and records failed fits separately from successful analyses.

Summary

This software constitutes a reproducible automated spectroscopy pipeline for quantitative analysis of ruby fluorescence spectra. Its modular structure separates data ingestion, preprocessing, model fitting, diagnostics and export, facilitating maintenance, validation and future extension.